

Day 36 - CDP & LLDP

Sunday, September 6, 2026

10:46 AM

Layer 2 Discovery Protocols Info

Layer 2 discovery protocols like Cisco Discovery Protocol (CDP) and Link Layer Discovery Protocol (LLDP) are used to share information and discover information about neighboring (directly connected) devices.

- **CDP** is a *Cisco proprietary* protocol
- **LLDP** is an *industry standard* protocol

The shared information can include Layer 2 and 3 data, like the *host name, IP address, device type*, etc.

They are usually considered to be security risks and are often not used in networks.

[Layer 2 Discovery Protocols - CDP and LLDP](#)

CDP and LLDP can both be run at the same time on the current networking device without any conflicts.

Cisco Discovery Protocol (CDP)

CDP Info

CDP is a *Cisco proprietary* protocol that is:

- Enabled globally by default on Cisco devices, and enabled by default on each interface
- Sends messages to **MAC Address: 0100.0ccc.cccc**
- **Timers:**
 1. **Message Timer: 60 seconds**
 2. **Holdtime: 180 seconds**
- **CDPv2** messages are sent by default

When a device receives a CDP message, it is processed and discarded afterwards. The device will NOT forward it to other devices.

CDP Show

- *show cdp*
 - Shows the global *message timer, holdtimer*, and **CDP version**
- *show cdp traffic*
 - Shows traffic statistics about **CDP** packets
- *show cdp interface*
 - Shows each interfaces' *encapsulation type, message timer, holdtimer*
 - Gives a summary of **CDP** enabled interfaces at the end
- *show cdp neighbors*
 - Shows a capability codes legend
 - Displays all the connected **CDP** devices along with their information, including: *Device ID, locally connected interface, current holdtime, capabilities, platform* and *port ID*
- *show cdp neighbors detail*
 - Shows detailed connected **CDP** devices information, including: All the previous information, *version, OS, CDP version, configured holdtime, VTP information, IP addresses of the neighboring connected interface, duplex mode, native VLAN*.
- *show cdp entry HOST-NAME*
 - Same as *show cdp neighbors detail*, but it shows information for the specified neighbor/host

CDP Config

- *cdp run*
 - Enables **CDP** globally
- *cdp enable*
 - Enables **CDP** on the current interface
- *cdp timer SECONDS*
 - Sets the **CDP** message *timer*
- *cdp holdtime SECONDS*

- Sets the **CDP** message *holdtime*
- *cdp advertise-v2*
 - Changes **CDP** version to **V2**

Link Layer Discovery Protocol (LLDP)

CDP Info

LLDP is an *industry standard* protocol that is:

- Disabled globally by default on Cisco devices, so it must be manually enabled and also enabled on each interface
- Sends messages to **MAC Address: 0180.c200.000e**
- **Timers:**
 1. **Message Timer:** *30 seconds*
 2. **Holdtime:** *120 seconds*
 3. **Reinitialization delay:** *2 seconds*
- **LLDP** has an *initialization timer (2 seconds)* that starts counting once **LLDP** is enabled (globally or on an interface), the timer will delay the actual initialization of **LLDP**.

When a device receives an LLDP message, it is processed and discarded afterwards. The device will NOT forward it to other devices.

LLDP Show

- *show lldp*
 - Shows the global *message timer*, *holdtimer*, and *initialization timer*
- *show lldp traffic*
 - Shows traffic statistics about **LLDP** packets
- *show lldp interface*
 - Shows each interfaces' *transmit*, *transmit state*, *receive* and *receive state*
- *show lldp neighbors*
 - Shows a capability codes legend
 - Displays all the connected **LLDP** devices along with their information, including: *Device ID*, *locally connected interface*, *configured holdtime*, *capabilities*, and *port ID*
- *show lldp neighbors detail*
 - Shows detailed connected **LLDP** devices information, including: All the previous information, *OS*, *current holdtime*, *system capabilities* and *enabled capabilities*
- *show lldp entry HOST-NAME*
 - Same as *show lldp neighbors detail*, but it shows information for the specified neighbor/host

CDP Config

- *lldp run*
 - Enables **LLDP** globally
- *lldp transmit*
 - Enables **LLDP** message transmission on the current interface
- *lldp receive*
 - Enables **LLDP** message receiving on the current interface
- *lldp timer SECONDS*
 - Sets the **LLDP** message *timer*
- *lldp holdtime SECONDS*
 - Sets the **LLDP** message *holdtime*
- *lldp reinit SECONDS*
 - Changes the **LLDP** *reinitialization timer*